

MULTIMODAL TRUST-AWARE BEHAVIORAL INTELLIGENCE SYSTEM USING MULTIMODAL AI, TEMPORAL REASONING, AND AGENTIC ANALYTICS

M.Sc

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2024 - 2026**

MULTIMODAL TRUST-AWARE BEHAVIORAL INTELLIGENCE SYSTEM USING MULTIMODAL AI, TEMPORAL REASONING, AND AGENTIC ANALYTICS

A thesis submitted in partial fulfillment of the requirements for the award of the
degree of

M.Sc.
in
AI & ML

by

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under the guidance of

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Indian Institute of Information Technology, Lucknow
2024-26

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AUTHORSHIP

I, Raghav Pimoli, declare that this thesis titled, "Multimodal trust-aware behavioral intelligence system using multimodal AI, temporal reasoning and agentic analytics" and the work presented in it are my own. I confirm that:

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Signed:

A square box containing a handwritten signature in blue ink that reads "Raghav".

Date:

30 May, 2026

CERTIFICATE

This is to certify that the thesis entitled “**Multimodal Trust-Aware Behavioral Intelligence System**” submitted by **Mr. Raghav Pimoli** who got his name registered on 20 Aug 2024 for the award of M.Sc degree at Indian Institute of Information Technology, Lucknow is absolutely based upon his own work under the supervision of **Dr. Sirsendu Sekhar Barman**, Department of Mathematics, and that neither this thesis nor any part of it has been submitted for any degree/diploma or any other academic award anywhere before.

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Acknowledgements

I would like to express my sincere gratitude to my supervisor **Dr. Sirsendu Sekhar Barman**, faculty members, and the Department of Information Technology at the Indian Institute of Information Technology, Lucknow for their guidance, encouragement, and continuous support throughout the development of this thesis.

I am especially thankful for the academic environment provided by the institute, which encouraged independent exploration, experimentation, and research-oriented thinking.

I would also like to thank my family and friends for their constant motivation and emotional support during the completion of this work.

Finally, I would like to acknowledge the open-source research community and the developers of PyTorch, HuggingFace Transformers, FastAPI, and Next.js, whose tools and frameworks contributed significantly to this project.

Lucknow
May, 2026

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ABSTRACT

The analysis of human behavior has emerged as a central challenge in modern artificial intelligence research, with wide-ranging implications for interview analytics, digital forensics, trust-aware decision support, behavioral monitoring, and human-computer interaction.

Conventional emotion recognition frameworks typically estimate discrete emotional states by relying on a single modality, such as facial imagery, acoustic signals, or linguistic content.

While such approaches have achieved competitive results on standard evaluation benchmarks, they are often inadequate when applied to realistic behavioral settings where emotional states shift continuously, appear ambiguous, or manifest differently across modalities.

This thesis introduces a Multimodal Trust-Aware Behavioral Intelligence System designed to surpass the limitations of standard emotion classifiers. Rather than predicting isolated emotional labels, the proposed system employs a layered behavioral reasoning architecture capable of examining behavioral consistency, the dynamics of trust over time, uncertainty progression, cross-modal disagreements, and long-term behavioral trajectories.

The system simultaneously processes three distinct modalities: visual information extracted from video recordings, acoustic characteristics derived from audio signals, and semantic content obtained from speech transcriptions. For visual processing, a MobileNetV2-based convolutional pipeline is employed. Acoustic features are represented using Mel-Frequency Cepstral Coefficients (MFCCs) and modeled using a Bidirectional Long Short-Term Memory (BiLSTM) network. Textual semantics are captured using DistilBERT, a compact transformer model. These three modality-specific representations are integrated through an attention-based fusion mechanism that learns to weight each modality according to its informativeness in a given context.

The framework introduces several novel components to address the limitations of prior work. These include an entropy-aware trust scoring mechanism, modules for temporal behavioral analysis, a graph-based engine for modeling behavioral state transitions, a memory-aware component for tracking persistent behavioral patterns, a cross-modal conflict detection module, and an agentic narrative reasoning engine powered by a large language model. Rather than producing numerical scores, the system generates structured behavioral intelligence reports in natural language, covering behavioral stability, uncertainty trends, trust dynamics, and cross-modal inconsistencies.

A full-stack behavioral analytics platform was also developed using Next.js and FastAPI. This platform supports real-time visualization, interactive behavioral analysis, and the automated generation of forensic PDF reports. Experimental evaluations demonstrate that the proposed architecture successfully identifies complex behavioral phenomena that standard multimodal classifiers typically overlook, establishing a foundation for future work in trust-aware and explainable behavioral intelligence.

Contents

1	Introduction	1
2	Literature Review	4
3	Methodology	6
4	Simulation and Results	9
5	Conclusion and Future work	17
6	Mathematical Formulation	18
7	References	19

List of Algorithms

Algorithm 1: Temporal Behavioral Analysis

Input: Segment-wise behavioral predictions

Output: Behavioral trends and volatility metrics

1. Extract per-segment confidence values from the classifier outputs.
2. Extract per-segment entropy values from the prediction distributions.
3. Compute trend slope across confidence and entropy sequences.
4. Compute behavioral volatility as the variance of successive prediction differences.
5. Return the computed temporal analytics for downstream use.

Algorithm 2: Graph-Based Behavioral Reasoning

6. Initialize the behavioral state graph with STABLE, UNCERTAIN, and INCONSISTENT nodes.
7. Map each segment's behavioral assessment to the corresponding graph state.
8. Compute transition counts and normalize to obtain transition probabilities.
9. Compute a stability score based on the fraction of time spent in the STABLE state.
10. Identify the dominant state and generate a behavioral trajectory interpretation.

Algorithm 3: Cross-Modal Conflict Detection

11. Extract the top predicted class from each modality independently.
12. Compare predictions across modalities to identify pairwise disagreements.
13. Count the number of modality pairs in disagreement for each segment.
14. Maintain a running record of conflict occurrences across the session.
15. Return a structured conflict analysis report.

Algorithm 4: Agentic Narrative Generation

16. Aggregate all behavioral analytics from preceding analysis layers.
17. Construct a structured reasoning context encoding temporal trends, trust scores, state transitions, and conflict summary.
18. Submit the constructed context to the large language model for narrative generation.
19. Return the generated natural-language behavioral intelligence report.

List of Figures

1	Confusion matrix	11
2	Behavioral Intelligence Dashboard	12
3	Trust Evolution Timeline	13
4	Behavioral State Timeline	14
5	Behavioral State Distribution	14
6	Cross Modal Agreement Analysis	14

List of Abbreviations & Notations

Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
BiLSTM	Bidirectional Long Short-Term Memory
NLP	Natural Language Processing
MFCC	Mel-Frequency Cepstral Coefficients
API	Application Programming Interface
GPU	Graphics Processing Unit
UI	User Interface
LLM	Large Language Model
FER	Facial Emotion Recognition

Chapter 1

INTRODUCTION

1.1 Background

Artificial intelligence has undergone significant transformation over the past decade, particularly in domains such as computer vision, natural language processing, speech analysis, and multimodal learning. Contemporary AI systems are no longer constrained to narrow, well-defined tasks. Rather, they are expected to operate across dynamic, real-world environments that demand a nuanced understanding of human behavior.

Human behavior is inherently multimodal. When people communicate, they simultaneously convey meaning through spoken language, facial expressions, vocal pitch and rhythm, physical posture, and temporal patterns of emotional change. Capturing and interpreting this rich behavioral signal is essential for applications including interview assessment, behavioral monitoring, mental health analytics, digital assistant design, deception detection, and trust-aware AI.

Traditional emotion recognition pipelines typically classify emotional states, such as happiness, sadness, fear, anger, or neutrality, from a single source of information. Although such systems can perform reasonably well in controlled experimental conditions, their reliability degrades in practical settings where emotional states are subtle, context-dependent, and continuously evolving.

Beyond predicting emotional labels, there exists a broader class of behavioral phenomena that current systems fail to address. These include behavioral consistency across time, uncertainty in model predictions, cross-modal contradiction, temporal drift, and the reliability of behavioral inferences. This thesis addresses these gaps by proposing a layered behavioral intelligence architecture that incorporates multimodal reasoning, trust-aware inference, temporal analysis, graph-based state modeling, and agentic narrative generation.

1.2 Problem Statement

Existing multimodal emotion recognition systems are constrained by several notable limitations:

- 1.They prioritize emotion classification over broader behavioral reasoning.
- 2.Behavioral analysis is typically performed on isolated segments rather than across time.
- 3.Most systems do not quantify prediction uncertainty or estimate trustworthiness.
- 4.Cross-modal contradictions between visual, acoustic, and linguistic channels are largely ignored.
- 5.Explainability remains limited in most deep learning-based approaches.
- 6.Memory-aware and agentic reasoning capabilities are absent from nearly all existing frameworks.
- 7.Most systems exist as offline research prototypes with no practical deployment infrastructure.

As a result, existing approaches fall short of the requirements imposed by real-world behavioral analytics applications.

1.3 Motivation

The motivation underlying this work stems from a simple but important observation: human behavior in practice is dynamic, context-dependent, and often contradictory across different modalities. A person may maintain composed facial expressions while exhibiting vocal stress indicators. Conversely, spoken words may carry positive sentiment while facial cues suggest discomfort. These cross-modal tensions carry significant behavioral information that isolated modality systems are structurally incapable of detecting.

This recognition gave rise to the goal of constructing a multimodal behavioral intelligence framework that could examine not just what a person appears to feel at a given moment, but how their behavior evolves, where inconsistencies arise, and whether the overall behavioral pattern is stable and trustworthy. The intent was never merely to improve emotion classification accuracy, but to build a layered AI architecture capable of genuine behavioral reasoning.

1.4 Objectives

The principal objectives of this thesis are as follows:

- 1.Design a multimodal AI architecture capable of processing video, audio, and text concurrently.
- 2.Implement an attention-based mechanism for multimodal feature fusion.
- 3.Develop a trust-aware scoring component for behavioral reliability estimation.
- 4.Analyze temporal behavioral consistency and detect patterns of instability.
- 5.Model behavioral state transitions using a graph-based reasoning approach.
- 6.Identify and quantify cross-modal behavioral conflicts.
- 7.Build an agentic reasoning layer that generates natural-language behavioral reports.
- 8.Deploy a behavioral intelligence platform with visualization and reporting capabilities.

1.5 Research Gaps

A thorough review of existing literature reveals several insufficiently addressed research challenges:

Research Gap 1: Limited Focus on Behavioral Intelligence

The majority of existing systems restrict their analysis to emotion classification, without examining broader questions of behavioral consistency or stability.

Research Gap 2: Lack of Temporal Behavioral Reasoning

Behavioral analysis in prior work is predominantly performed on isolated clips. Long-term behavioral evolution across time is rarely studied.

Research Gap 3: Absence of Trust-Aware Reasoning

Most systems report class probabilities without assessing whether those predictions are reliable or uncertain.

Research Gap 4: No Graph-Based Behavioral Modeling

Graph-based methods for modeling transitions between behavioral states are conspicuously absent from existing multimodal frameworks.

Research Gap 5: Weak Explainability

Deep learning models in this domain generally function as black boxes, offering limited interpretability.

Research Gap 6: Lack of Cross-Modal Conflict Analysis

Despite the natural possibility of disagreement between modalities, most fusion approaches do not model or report such contradictions.

Research Gap 7: Absence of Agentic Behavioral Reasoning

Higher-order reasoning about behavior, including the generation of interpretable narrative summaries, is absent from nearly all existing systems.

1.6 Proposed Solution

To address the gaps identified above, this thesis proposes a layered behavioral intelligence architecture comprising nine principal components:

1. Multimodal perception layer
2. Attention-based fusion layer
3. Trust-aware scoring layer
4. Temporal behavioral reasoning module
5. Graph-based state reasoning engine
6. Behavioral memory and persistence layer
7. Cross-modal conflict detection module
8. Agentic narrative reasoning engine
9. Frontend behavioral analytics platform

These components are unified within a single framework that integrates deep learning, temporal analytics, graph reasoning, explainable AI, and large language model-based interpretation.

1.7 Major Contributions

The key contributions of this thesis are summarized as follows:

10. A multimodal behavioral intelligence architecture that operates across video, audio, and text.
11. An entropy-aware trust scoring mechanism for behavioral reliability estimation.
12. A temporal behavioral reasoning module for tracking evolving behavioral patterns.
13. A graph-based engine for modeling transitions between behavioral states.
14. A cross-modal conflict detection and reasoning module.
15. A memory-aware component for tracking persistent behavioral patterns.
16. An agentic narrative intelligence layer capable of producing human-readable behavioral reports.
17. A complete full-stack behavioral analytics platform with visualization capabilities.
18. An automated forensic PDF report generation pipeline.

1.8 Thesis Organization

The remainder of this thesis is structured as follows. Chapter 2 reviews the relevant literature and surveys existing systems. Chapter 3 presents the proposed methodology and system architecture in detail. Chapter 4 describes the implementation, experimental setup, and key observations. Chapter 5 concludes the thesis and outlines directions for future research.

Chapter 2

Literature Review

2.1 Introduction

Progress in multimodal AI has accelerated considerably in recent years, driven by advancements in deep learning, computer vision, natural language processing, and acoustic signal analysis. A growing body of research has explored the combination of facial expressions, speech signals, and linguistic content for the purpose of emotion recognition. Notwithstanding these advances, the field continues to exhibit significant gaps in behavioral reasoning, temporal modeling, and trust-aware analysis.

2.2 Traditional Emotion Recognition Systems

Early approaches to emotion recognition relied on manually engineered features. Common examples include facial landmark coordinates, Mel-Frequency Cepstral Coefficients (MFCCs), and various statistical signal descriptors. While these methods produced acceptable results in structured laboratory environments, they were poorly suited to real-world deployment. Challenges included sensitivity to environmental noise, difficulty handling temporal variability, and an inability to reconcile conflicting signals across modalities.

2.3 Deep Learning-Based Emotion Recognition

The widespread adoption of convolutional neural networks (CNNs) and recurrent neural networks (RNNs) substantially improved the performance of emotion recognition systems. CNN architectures became the standard approach for visual emotion analysis, while recurrent models, particularly LSTM and BiLSTM networks, demonstrated strong performance on speech-based tasks. Transformer architectures subsequently enhanced the contextual understanding of language for sentiment and emotion prediction.

Despite these improvements, deep learning-based emotion recognition systems continue to exhibit several weaknesses. They often overfit to narrow benchmark datasets, provide insufficient explanations for their predictions, lack mechanisms for assessing prediction trustworthiness, and do not engage in broader behavioral reasoning.

2.4 Multimodal Fusion Approaches

Multimodal fusion attempts to exploit complementary information from multiple signal sources to produce more robust and accurate predictions. The dominant fusion paradigms include early fusion, which combines raw features before classification; late fusion, which merges independent modality predictions; and hybrid approaches that integrate both strategies.

Although multimodal fusion generally improves robustness, existing systems tend to treat modalities as static, independent sources of information rather than as components of a coherent behavioral signal. Explicit reasoning about cross-modal disagreement, which can itself serve as a meaningful behavioral indicator, is rarely addressed.

2.5 Explainable AI and Trust-Aware Systems

The need for interpretable and trustworthy AI has gained substantial traction within the research community. It is now widely recognized that prediction outputs alone are inadequate for high-stakes applications. Researchers have explored confidence calibration, entropy-based uncertainty estimation, and reliability-aware prediction frameworks. However, these ideas have rarely been adapted for use within multimodal behavioral analysis pipelines, leaving a significant opportunity for integration.

2.6 Research Gap Summary

On the basis of the literature reviewed, the following challenges remain insufficiently addressed in existing work:

- Systematic analysis of behavioral consistency across modalities and time
- Temporal modeling of behavioral evolution
- Graph-based reasoning about behavioral state transitions
- Trust-aware inference with uncertainty quantification
- Detection and analysis of cross-modal contradictions
- Memory-aware behavioral persistence modeling
- Natural-language narrative explainability
- Deployable behavioral intelligence platforms

The proposed thesis targets each of these gaps through a unified, layered behavioral intelligence framework.

Chapter 3

Methodology

3.1 System Overview

The proposed system is a layered multimodal behavioral intelligence framework designed to analyze human behavior from three concurrent input streams: video, audio, and transcribed text. The overall processing pipeline consists of the following stages:

- 1.Temporal video segmentation
- 2.Modality-specific preprocessing
- 3.Deep feature extraction per modality
- 4.Attention-based multimodal fusion
- 5.Trust-aware behavioral scoring
- 6.Temporal behavioral reasoning
- 7.Graph-based state reasoning
- 8.Cross-modal conflict detection
- 9.Agentic narrative generation
10. Frontend visualization and reporting

3.2 Dataset Preparation

IEMOCAP Dataset

The Interactive Emotional Dyadic Motion Capture (IEMOCAP) dataset was used to provide broad emotional coverage across multiple speakers and interaction scenarios. Its multimodal nature, encompassing video, audio, and transcripts, made it a suitable foundation for initial model training.

Custom Behavioral Dataset

To extend the system's capability to realistic interview scenarios, fine-tuning was additionally performed using a purpose-built dataset. This dataset included recordings of baseline behavioral patterns, scripted verbal responses, time-pressured interactions, and intentionally varied behavioral conditions. This two-stage training strategy improved the model's ability to generalize beyond the controlled conditions represented in benchmark corpora.

3.3 Temporal Video Segmentation

Input videos are divided into fixed-duration temporal segments prior to processing. Each segment is analyzed independently across all three modalities, after which segment-level results are aggregated for temporal reasoning. This segmentation strategy allows the system to track how behavior shifts over time, enabling the analysis of behavioral evolution rather than static emotional snapshots.

3.4 Video Modality Processing

Visual features are extracted using MobileNetV2, a lightweight convolutional neural network originally designed for mobile deployment. Individual frames are sampled from each video segment and processed through the convolutional backbone. Temporal averaging is then applied across frames within each segment to yield stable visual embeddings. The visual processing pipeline is sensitive to facial behavior, subtle postural cues, motion patterns, and visually expressed emotional signals.

3.5 Audio Modality Processing

Audio signals corresponding to each video segment are extracted and transformed into MFCC representations, which encode the spectral envelope of the speech signal in a compact, perceptually meaningful form. These features are modeled using a Bidirectional LSTM network, which processes the temporal sequence in both forward and backward directions. This architecture is well suited to capturing vocal dynamics including pitch variation, intensity shifts, speaking rhythm, and patterns associated with vocal stress.

3.6 Text Modality Processing

Speech content from each audio segment is converted to text through automatic transcription. The resulting text is processed using DistilBERT, a knowledge-distilled variant of the BERT transformer that retains strong contextual representation capabilities while reducing computational overhead. The text modality contributes semantic intent, linguistic tone, and context-sensitive emotional cues that complement the visual and acoustic streams.

3.7 Attention-Based Fusion

A central contribution of this architecture is its use of an attention-based mechanism for multimodal integration. Unlike simple feature concatenation, which treats all modalities as equally informative, the attention mechanism learns to assign dynamic weights to each modality depending on the behavioral context. This allows the system to reduce the influence of an unreliable or low-quality modality in situations where, for instance, acoustic quality is degraded or facial visibility is limited.

3.8 Trust Scoring Layer

The trust scoring layer quantifies the reliability of behavioral predictions by combining several complementary signals. These include prediction confidence, entropy-normalized uncertainty, temporal prediction volatility, and the rate of abrupt prediction changes across consecutive segments. This formulation allows the system to distinguish between confident and stable behavioral estimates on the one hand, and unreliable or fluctuating predictions on the other.

3.9 Temporal Behavioral Reasoning

Temporal reasoning modules examine how behavioral indicators evolve across the sequence of analyzed segments. The system monitors confidence trends, entropy trajectories, behavioral drift between consecutive predictions, and segment-level prediction volatility. These measurements collectively support the identification of deteriorating behavioral stability, progressive uncertainty, and other temporally defined patterns.

3.10 Graph-Based Behavioral Reasoning

Behavioral states are modeled as nodes in a graph, with edges representing transitions between states over time. Three primary states are defined: STABLE, UNCERTAIN, and INCONSISTENT. The graph reasoning engine computes state transition probabilities, identifies dominant behavioral states, detects recovery from instability, and characterizes the overall trajectory of behavioral dynamics. This layer provides a richer and more structured interpretation of behavioral evolution than raw classification outputs alone could offer.

3.11 Cross-Modal Conflict Reasoning

This module assesses the degree of agreement between the predictions generated independently by each modality. Persistent disagreement between the video, audio, and text predictions is treated as evidence of behavioral inconsistency and flagged accordingly. This novel component introduces an explicit measure of multimodal alignment, which can serve as an independent behavioral signal in its own right.

3.12 Behavioral Memory Layer

A memory-aware behavioral manager maintains a structured record of behavioral observations across the full duration of an analyzed session. The memory layer tracks repeated episodes of uncertainty, extended periods of instability, and sustained cross-modal conflicts. This historical context enables the system to distinguish isolated anomalous events from persistent behavioral patterns, supporting more contextually grounded inference.

3.13 Agentic Narrative Reasoning

A large language model serves as the final reasoning layer, transforming the structured numerical outputs of the preceding components into coherent natural-language behavioral reports. These reports synthesize information about behavioral consistency, trust evolution, uncertainty trends, modal agreement, and overall behavioral assessment. The narrative output is designed to be accessible to non-expert stakeholders while remaining technically grounded.

3.14 Frontend Behavioral Intelligence Platform

A complete web-based analytics platform was developed to support practical deployment of the system. The backend was implemented using FastAPI, and the frontend was built using Next.js with TailwindCSS for visualization. The platform provides interactive trust evolution graphs, behavioral state timelines, suspicious segment heatmaps, narrative behavioral reports, and one-click forensic PDF export.

Chapter 4

Simulation and Results

4.1 Introduction

This chapter presents the experimental evaluation of the proposed Multimodal Trust-Aware Behavioral Intelligence System. The primary objective of the experimentation was to assess not only the classification capability of the multimodal fusion architecture but also the effectiveness of the additional reasoning layers introduced in this work, including trust-aware analytics, temporal behavioral analysis, graph-based state reasoning, cross-modal consistency analysis, and agentic narrative intelligence.

Unlike conventional emotion recognition systems that focus solely on classification accuracy, the proposed framework aims to provide a richer understanding of behavioral dynamics over time. Therefore, the evaluation process includes both traditional machine learning metrics and behavioral intelligence indicators generated by the system.

The experiments were conducted using validation samples derived from the behavioral dataset after model fine-tuning. All evaluations were performed using the complete multimodal pipeline consisting of video, audio, and textual inputs.

4.2 Experimental Environment

The system was developed and evaluated using the hardware and software configuration shown in Table 4.1.

Table 4.1 Experimental Environment

Component	Specification
GPU	NVIDIA RTX 3050
Deep Learning Framework	PyTorch
NLP Framework	HuggingFace Transformers
Backend Framework	FastAPI
Frontend Framework	Next.js
Video Backbone	MobileNetV2
Audio Model	BiLSTM
Text Model	DistilBERT
Programming Language	Python

The complete multimodal inference pipeline was integrated into a web-based behavioral intelligence platform capable of performing end-to-end behavioral analysis from uploaded interview videos.

4.3 Overall Classification Performance

The performance of the multimodal fusion model was evaluated using standard classification metrics including accuracy, precision, recall, and F1-score.

Table 4.2 Overall Validation Metrics

Metric	Value
Accuracy	70.83%
Precision	81.34%
Recall	70.83%
F1-Score	70.41%

The obtained results indicate that the proposed multimodal architecture was able to learn meaningful representations from heterogeneous behavioral inputs.

An overall validation accuracy of 70.83% demonstrates that the model correctly identified behavioral categories in a majority of validation samples. Although the classification problem remains challenging due to overlapping emotional and behavioral characteristics, the achieved performance confirms the effectiveness of multimodal learning.

A noteworthy observation is the relatively high weighted precision of 81.34%. This suggests that when the model assigned a behavioral label, the prediction was generally reliable and associated with comparatively few false positive classifications. From a behavioral intelligence perspective, maintaining high precision is important because incorrect behavioral interpretations can negatively influence subsequent reasoning stages.

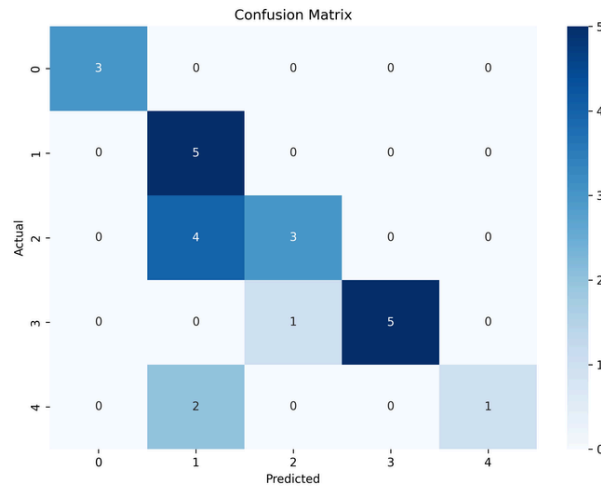
The weighted recall of 70.83% indicates that certain behavioral categories remained difficult to detect consistently. This behavior is expected in multimodal behavioral analysis because several emotional states may exhibit similar facial expressions, speech characteristics, or linguistic patterns.

The resulting F1-score of 70.41% demonstrates a balanced trade-off between prediction reliability and class coverage. Considering the complexity of the task and the multimodal nature of the data, these results provide evidence that the fusion architecture successfully captured useful information from all three modalities.

4.4 Confusion Matrix Analysis

To better understand model behavior, a confusion matrix was generated using validation predictions.

Figure 4.1 Confusion Matrix of the Multimodal Fusion Model



The confusion matrix provides a detailed view of how different behavioral classes were recognized by the model and highlights areas where misclassification occurred.

Several observations can be drawn from the matrix.

Class 0 achieved perfect classification performance, with all validation samples correctly identified. Similarly, Class 1 achieved complete recall, indicating that every sample belonging to this category was successfully detected.

Class 3 also demonstrated strong performance, with the majority of samples being classified correctly. This suggests that the multimodal characteristics associated with these behavioral categories were sufficiently distinctive for the model to learn robust decision boundaries. The majority of classification errors occurred between Classes 1, 2, and 4. In particular, a considerable number of Class 2 samples were assigned to Class 1. This behavior suggests that these categories share overlapping behavioral signatures that are difficult to separate completely using multimodal features alone.

A similar trend was observed for Class 4, where some instances were classified as Class 1. Such confusion is understandable because behavioral states rarely exist as completely independent categories. Human behavior often exhibits blended emotional characteristics, making strict separation inherently challenging.

These findings demonstrate the limitations of relying solely on categorical predictions and further justify the inclusion of higher-level reasoning modules within the proposed framework.

4.5 Class-Wise Performance Analysis

To obtain deeper insights into model behavior, class-wise precision, recall, and F1-scores were analyzed.

Table 4.3 Class-Wise Evaluation Results

Class	Precision	Recall	F1-Score
0	1.00	1.00	1.00
1	0.45	1.00	0.62
2	0.75	0.43	0.55
3	1.00	0.83	0.91
4	1.00	0.33	0.50

The class-wise analysis reveals significant variation across behavioral categories.

Classes 0 and 3 exhibited the strongest overall performance. Both classes achieved high precision and recall values, indicating that their behavioral patterns were consistently distinguishable across modalities.

Class 1 achieved perfect recall but comparatively lower precision. This indicates that while all genuine samples belonging to Class 1 were detected successfully, samples from other categories were occasionally assigned to this class. Consequently, Class 1 appears to occupy a broader decision region within the learned feature space.

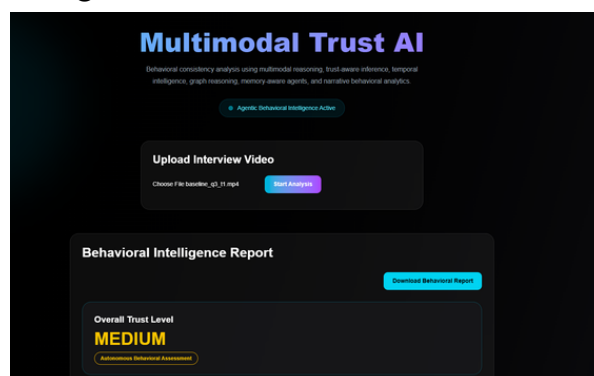
Classes 2 and 4 presented greater challenges for the model. Although precision remained relatively high, recall values were noticeably lower. This suggests that several samples belonging to these categories were confused with neighboring behavioral states.

The observed performance variation highlights the complexity of multimodal behavioral understanding and demonstrates that certain behavioral states are inherently more difficult to distinguish than others.

4.6 Behavioral Intelligence Dashboard Evaluation

A key contribution of this research is the development of an integrated Behavioral Intelligence Dashboard capable of visualizing behavioral dynamics beyond traditional classification outputs.

Figure 4.2 Behavioral Intelligence Dashboard



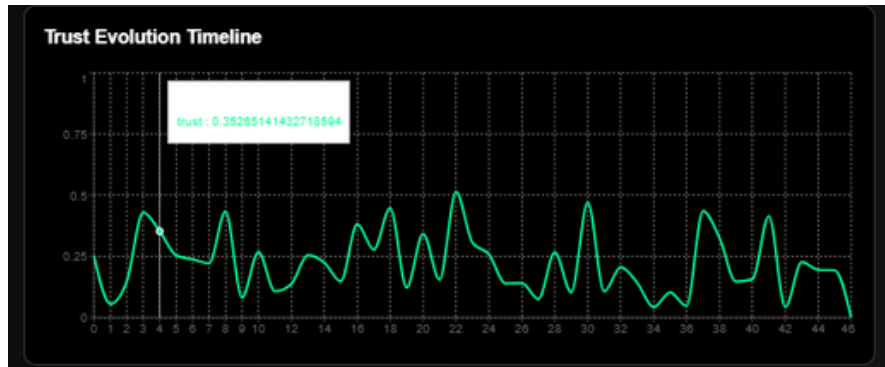
The dashboard combines outputs from multiple analytical layers and presents them through an interactive interface. The generated behavioral report includes trust evolution analysis, behavioral state tracking, cross-modal consistency evaluation, suspicious segment identification, and agentic narrative reasoning.

This additional interpretability layer significantly enhances the practical usability of the system. Rather than presenting isolated emotion predictions, the dashboard provides a holistic view of behavioral evolution throughout an interview session.

4.7 Trust Evolution Analysis

The trust evolution module tracks confidence and uncertainty across temporal segments generated during inference.

Figure 4.3 Trust Evolution Timeline



Analysis of the generated trust timeline revealed noticeable fluctuations throughout the evaluated interview session.

The trust values did not remain constant across the interaction, indicating varying levels of behavioral consistency. Multiple peaks and troughs were observed, suggesting periods of relatively stable behavior followed by intervals of increased uncertainty.

The overall trust assessment generated by the system was categorized as Medium. This result indicates that while the behavioral evidence was generally reliable, certain segments exhibited fluctuations that reduced overall confidence.

An important observation is that trust degradation occurred gradually rather than abruptly. This pattern suggests that uncertainty accumulated progressively over time instead of being triggered by isolated events.

Such temporal information would not be visible through conventional classification metrics alone and demonstrates the value of trust-aware behavioral reasoning.

4.8 Behavioral State Analysis

Behavioral states were categorized into three primary classes:

- Stable
- Uncertain
- Inconsistent

Figure 4.4 Behavioral State Timeline

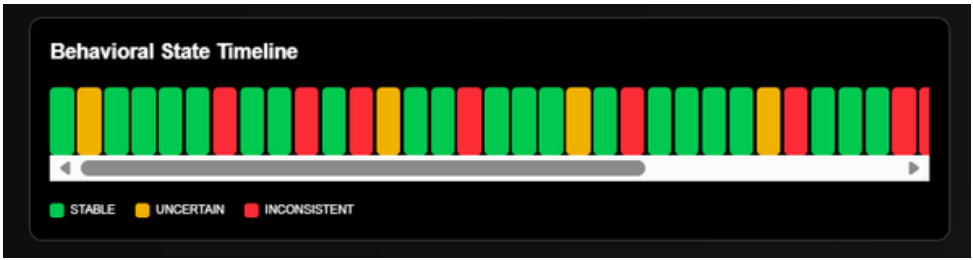
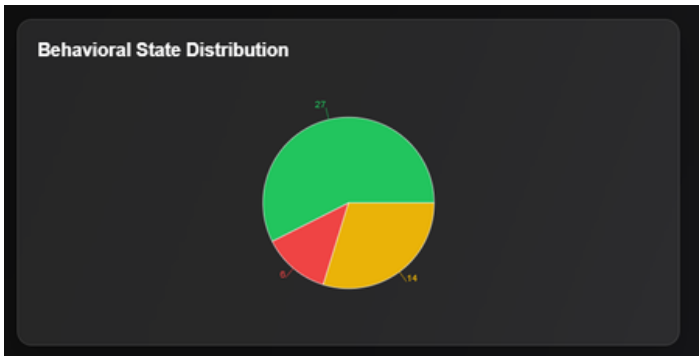


Figure 4.5 Behavioral State Distribution



The behavioral state distribution generated by the system is summarized below.

Table 4.4 Behavioral State Distribution

State	Count
Stable	27
Uncertain	14
Inconsistent	6

The results indicate that stable behavior dominated the analyzed session. More than half of the observed segments were categorized as stable, suggesting a generally consistent behavioral pattern.

However, the presence of uncertain and inconsistent segments indicates intermittent fluctuations throughout the interaction. These fluctuations were distributed across the timeline rather than being concentrated within a single region, suggesting localized behavioral variations rather than persistent instability.

The state distribution further supports the medium trust assessment generated by the system.

4.9 Cross-Modal Agreement Analysis

The proposed framework evaluates consistency between video, audio, and text predictions before generating final behavioral interpretations.

Figure 4.6 Cross-Modal Agreement Analysis

Cross-Modal Agreement Analysis					
Segment	Audio	Text	Video	Fusion	Agreement
0				ANGRY	CONSISTENT
1				ANGRY	CONSISTENT
2				ANGRY	CONSISTENT
3				FEAR	CONSISTENT
4				FEAR	CONSISTENT
5				FEAR	CONSISTENT
6				NEUTRAL	CONSISTENT
7				ANGRY	CONSISTENT
8				ANGRY	CONSISTENT
9				FEAR	CONSISTENT
10				FEAR	CONSISTENT
11				ANGRY	CONSISTENT
12				ANGRY	CONSISTENT

The evaluated interview session exhibited a high degree of multimodal agreement. Most temporal segments demonstrated consistent behavioral predictions across modalities, resulting in stable fusion outputs.

No persistent modal conflicts were detected during the analyzed session.

This observation suggests that the behavioral inconsistencies observed by the system were primarily caused by temporal fluctuations rather than contradictory evidence from different modalities.

The absence of major modal disagreements increases confidence in the final behavioral interpretations generated by the reasoning engine.

4.10 Agentic Behavioral Intelligence Analysis

One of the most distinctive aspects of the proposed architecture is the introduction of an Agentic Behavioral Intelligence Layer.

The reasoning engine generated the behavioral metrics summarized in Table 4.5.

Table 4.5 Agentic Behavioral Intelligence Metrics

Metric	Value
Stability Score	0.43
Flip Rate	0.47
Instability Score	0.04
Persistent Uncertain Runs	2
Persistent Inconsistent Runs	2
Dominant Behavioral State	Stable
Modal Conflicts	0

The dominant behavioral state was identified as Stable. Nevertheless, the stability score remained moderate, indicating that the behavioral pattern was not entirely uniform.

The flip rate of 0.47 suggests that behavioral predictions changed frequently across temporal segments. This finding aligns with the trust evolution analysis, which revealed continuous fluctuations in confidence.

The instability score remained relatively low, implying that behavioral variations were moderate rather than extreme. Furthermore, no escalation patterns were detected, indicating the absence of rapidly deteriorating behavioral conditions.

The generated narrative report concluded that the subject exhibited moderate behavioral consistency with signs of gradual confidence reduction and increasing uncertainty over time. Such conclusions would be difficult to obtain from traditional classification systems that rely solely on categorical outputs.

4.11 Discussion

The experimental results demonstrate that the proposed framework extends beyond conventional multimodal emotion recognition systems.

While the classification performance itself is encouraging, the most significant contribution lies in the additional reasoning capabilities introduced through trust-aware analytics, temporal intelligence, graph-based behavioral modeling, and agentic interpretation.

The behavioral intelligence modules successfully transformed raw prediction outputs into meaningful behavioral insights, enabling deeper analysis of consistency, uncertainty, stability, and behavioral evolution.

The integration of these reasoning layers allows the system to move beyond simple emotion recognition and toward a more comprehensive behavioral intelligence paradigm.

Overall, the findings indicate that the proposed architecture is capable of supporting real-world behavioral analysis applications where interpretability and trustworthiness are as important as predictive performance.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This thesis presented a Multimodal Trust-Aware Behavioral Intelligence System capable of analyzing behavioral consistency, uncertainty progression, trust dynamics, and cross-modal conflicts using a layered AI architecture. By going beyond conventional emotion classification, the system introduces temporal reasoning, graph-based state modeling, entropy-aware trust scoring, memory-based behavioral persistence analysis, and large language model-driven narrative generation.

Experimental results demonstrated that the proposed architecture captures complex behavioral phenomena, including temporal drift, state instability, and cross-modal disagreement, that standard multimodal classifiers routinely miss. The accompanying full-stack behavioral analytics platform further enhances the system's practical value by enabling interactive visualization, comprehensive behavioral reporting, and automated forensic document generation.

The work contributes a substantive and extensible framework to the growing field of trust-aware multimodal behavioral AI, providing both a technical foundation and a practical template for future research and deployment.

5.2 Future Work

Several directions exist for extending the contributions of this thesis:

Real-Time Streaming Analysis

Future work may adapt the system for live interview or behavioral monitoring scenarios, requiring low-latency inference and real-time reporting pipelines.

Transformer-Based Multimodal Fusion

Large-scale multimodal transformer architectures may replace the current attention-based fusion module, potentially improving representational richness and cross-modal alignment.

Reinforcement Learning-Based Behavioral Agents

The agentic reasoning layer could be enhanced using reinforcement learning-based agents capable of adaptive decision-making and personalized behavioral modeling.

Physiological Signal Integration

The inclusion of additional physiological modalities, such as electroencephalography (EEG), electrocardiography (ECG), or eye-tracking data, may substantially enrich the system's behavioral understanding.

Large-Scale Behavioral Datasets

Collecting and utilizing larger, more diverse behavioral datasets would improve the generalizability of the trained models.

Edge Deployment

Optimization of the inference pipeline for deployment on edge devices would expand the range of practical applications and reduce dependence on cloud computing infrastructure.

CHAPTER 6

MATHEMATICAL FORMULATION

6.1 Multimodal Fusion Formulation

Let V , A , and T denote the feature embeddings extracted from the video, audio, and text modalities respectively. The fused multimodal representation is formed by concatenating these embeddings:

$$F = [V \parallel A \parallel T]$$

where $\parallel$ denotes feature concatenation. The classification layer then computes predicted behavioral probabilities as:

$$\hat{y} = \text{Softmax}(WF + b)$$

where W represents the trainable weight matrix and b denotes the bias vector.

6.2 Trust Score Formulation

The behavioral trust score is computed as a linear combination of prediction confidence, entropy, volatility, and prediction flip rate:

$$\text{Trust} = C - \alpha E - \beta V - \gamma F$$

where C denotes prediction confidence, E denotes entropy, V denotes behavioral volatility, and F denotes the prediction flip rate. The coefficients α , β , and γ are tunable hyperparameters.

6.3 Entropy Formulation

Behavioral uncertainty is quantified using Shannon entropy:

$$H(p) = - \sum p_i \log(p_i)$$

Higher entropy values indicate greater prediction uncertainty and reduced behavioral stability.

6.4 Behavioral Stability Score

Behavioral stability is estimated from the frequency of state transitions:

$$S = 1 - (F_t / N)$$

where F_t denotes the number of behavioral state transitions and N denotes the total number of analyzed segments. Values approaching one indicate high stability.

6.5 Cross-Modal Conflict Score

The cross-modal conflict score is defined as:

$$CM = M_n / M_t$$

where M_n is the number of modalities producing conflicting predictions and M_t is the total number of modalities under consideration.

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